CSE 242 – Assignment 1

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Part 1: Linear Algebra

Q1

$A = \begin{bmatrix} 3 & 4 \\ 2 & 1 \end{bmatrix}, B = \begin{bmatrix} 1 & 2 \\ 0 & -1 \end{bmatrix}$.

$$AB = \begin{bmatrix} 3\cdot 1 + 4\cdot 0 & 3\cdot 2 + 4\cdot(-1) \\ 2\cdot 1 + 1\cdot 0 & 2\cdot 2 + 1\cdot(-1) \end{bmatrix} = \begin{bmatrix} 3 & 2 \\ 2 & 3 \end{bmatrix}$$

$$BA = \begin{bmatrix} 1\cdot 3 + 2\cdot 2 & 1\cdot 4 + 2\cdot 1 \\ 0\cdot 3 + (-1)\cdot 2 & 0\cdot 4 + (-1)\cdot 1 \end{bmatrix} = \begin{bmatrix} 7 & 6 \\ -2 & -1 \end{bmatrix}$$

Note $AB \neq BA$: matrix multiplication is not commutative.

$$A + B = \begin{bmatrix} 4 & 6 \\ 2 & 0 \end{bmatrix} \Rightarrow (A+B)^T = \begin{bmatrix} 4 & 2 \\ 6 & 0 \end{bmatrix}$$

$$B^T = \begin{bmatrix} 1 & 0 \\ 2 & -1 \end{bmatrix}, \quad AB^T = \begin{bmatrix} 3+8 & 0-4 \\ 2+2 & 0-1 \end{bmatrix} = \begin{bmatrix} 11 & -4 \\ 4 & -1 \end{bmatrix}$$

$$\operatorname{trace}(A) = 3 + 1 = 4, \qquad \operatorname{trace}(AB) = 3 + 3 = 6.$$

(Also $\operatorname{trace}(BA) = 7 - 1 = 6$, consistent with Q3.)

Q2

For $M = \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}$, the characteristic polynomial is

$$\det(M - \lambda I) = (2-\lambda)^2 - 1 = \lambda^2 - 4\lambda + 3 = (\lambda - 1)(\lambda - 3) = 0,$$

so $\lambda_1 = 1$ and $\lambda_2 = 3$.

$\lambda = 1$: $(M - I)v = \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} v = 0 \Rightarrow v_1 = v_2$, so $v = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$. Check: $Mv = \begin{bmatrix} 2-1 \\ -1+2 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix} = 1 \cdot v$.

$\lambda = 3$: $(M - 3I)v = \begin{bmatrix} -1 & -1 \\ -1 & -1 \end{bmatrix} v = 0 \Rightarrow v_1 = -v_2$, so $v = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$. Check: $Mv = \begin{bmatrix} 2+1 \\ -1-2 \end{bmatrix} = \begin{bmatrix} 3 \\ -3 \end{bmatrix} = 3 \cdot v$.

Normalized eigenvectors are $\frac{1}{\sqrt{2}}[1, 1]^T$ and $\frac{1}{\sqrt{2}}[1, -1]^T$. They are orthogonal, as expected for a symmetric matrix.

Q3

Let A be $m \times n$ and B be $n \times m$. By definition $(AB)_{ii} = \sum_{k=1}^{n} A_{ik} B_{ki}$, so

$$\operatorname{trace}(AB) = \sum_{i=1}^{m} \sum_{k=1}^{n} A_{ik} B_{ki} = \sum_{k=1}^{n} \sum_{i=1}^{m} B_{ki} A_{ik} = \sum_{k=1}^{n} (BA)_{kk} = \operatorname{trace}(BA),$$

where the sums can be exchanged because they are finite, and $A_{ik} B_{ki} = B_{ki} A_{ik}$ since scalars commute. ∎

Q4

$C = \begin{bmatrix} 4 & 7 \\ 2 & 6 \end{bmatrix}$. **Existence:** $\det C = 4 \cdot 6 - 7 \cdot 2 = 24 - 14 = 10 \neq 0$, so C is invertible.

Inverse: for $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$, the inverse is $\frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$:

$$C^{-1} = \frac{1}{10}\begin{bmatrix} 6 & -7 \\ -2 & 4 \end{bmatrix} = \begin{bmatrix} 0.6 & -0.7 \\ -0.2 & 0.4 \end{bmatrix}.$$

Check: $CC^{-1} = \frac{1}{10}\begin{bmatrix} 24-14 & -28+28 \\ 12-12 & -14+24 \end{bmatrix} = I$.

Part 2: Probability, MLE, and MAP Estimation

Q1

(a) The probabilities must sum to one. Using $e^{x} = \sum_{k \ge 0} x^k/k!$,

$$1 = c\sum_{k=0}^{\infty} \frac{3^k}{k!} = c\,e^{3} \;\Rightarrow\; c = e^{-3} \approx 0.0498.$$

So $X \sim \text{Poisson}(3)$.

(b)

$$E[X] = \sum_{k=1}^{\infty} k\,e^{-3}\frac{3^k}{k!} = 3e^{-3}\sum_{k=1}^{\infty}\frac{3^{k-1}}{(k-1)!} = 3e^{-3}e^{3} = 3.$$

$$E[X(X-1)] = \sum_{k=2}^{\infty} k(k-1)\,e^{-3}\frac{3^k}{k!} = 9e^{-3}\sum_{k=2}^{\infty}\frac{3^{k-2}}{(k-2)!} = 9.$$

$$E[X^2] = E[X(X-1)] + E[X] = 12, \qquad \text{Var}(X) = E[X^2] - E[X]^2 = 12 - 9 = 3.$$

Q2

(a) $P(A \cap B) = P(A \mid B)\,P(B) = 0.6 \times 0.3 = 0.18$.

(b) $P(B \mid A) = \dfrac{P(A \cap B)}{P(A)} = \dfrac{0.18}{0.5} = 0.36$.

(c) $P(A \cup B) = P(A) + P(B) - P(A \cap B) = 0.5 + 0.3 - 0.18 = 0.62$.

Q3

The sequence H, H, T, H, T has $h = 3$ heads and $t = 2$ tails in $n = 5$ independent flips.

(a) MLE. The likelihood is $L(\theta) = \prod_i P(x_i \mid \theta) = \theta^3(1-\theta)^2$, so

$$\ell(\theta) = 3\ln\theta + 2\ln(1-\theta), \qquad \frac{d\ell}{d\theta} = \frac{3}{\theta} - \frac{2}{1-\theta} = 0 \;\Rightarrow\; 3(1-\theta) = 2\theta \;\Rightarrow\; \hat{\theta}_{\text{MLE}} = \frac{3}{5} = 0.6.$$

Since $\ell''(\theta) = -\frac{3}{\theta^2} - \frac{2}{(1-\theta)^2} < 0$, this is a maximum.

(b) MAP. With prior $P(\theta) = 2\theta$ on $[0,1]$,

$$P(\theta \mid D) \propto \theta^3(1-\theta)^2 \cdot 2\theta \propto \theta^4(1-\theta)^2.$$

$$\frac{d}{d\theta}\left[4\ln\theta + 2\ln(1-\theta)\right] = \frac{4}{\theta} - \frac{2}{1-\theta} = 0 \;\Rightarrow\; 4(1-\theta) = 2\theta \;\Rightarrow\; \hat{\theta}_{\text{MAP}} = \frac{2}{3} \approx 0.667.$$

The second derivative is again negative, so this is a maximum. The prior 2θ is a $\text{Beta}(2,1)$ density, which acts like one extra observed head and pulls the estimate up from 0.6. In general $\hat{\theta}_{\text{MAP}} = \frac{h+1}{n+1}$. When $h = n$ the posterior θ^{n+1} is increasing and the maximum is at the boundary $\theta = 1$, which agrees with the formula.

Q4

For i.i.d. $x_1, \dots, x_N \sim \mathcal{N}(\mu, \sigma^2)$, the log-likelihood is

$$\ell(\mu, \sigma^2) = -\frac{N}{2}\ln(2\pi) - \frac{N}{2}\ln\sigma^2 - \frac{1}{2\sigma^2}\sum_{i=1}^{N}(x_i - \mu)^2.$$

$$\frac{\partial \ell}{\partial \mu} = \frac{1}{\sigma^2}\sum_i (x_i - \mu) = 0 \;\Rightarrow\; \hat{\mu} = \frac{1}{N}\sum_i x_i.$$

$$\frac{\partial \ell}{\partial \sigma^2} = -\frac{N}{2\sigma^2} + \frac{1}{2\sigma^4}\sum_i (x_i - \mu)^2 = 0 \;\Rightarrow\; \hat{\sigma}^2 = \frac{1}{N}\sum_i (x_i - \hat{\mu})^2.$$

For $X = \{1, 3, 5, 7\}$, $N = 4$: $\hat{\mu} = \frac{16}{4} = 4$. The deviations are $-3, -1, 1, 3$ with squares $9, 1, 1, 9$, so $\hat{\sigma}^2 = \frac{20}{4} = 5$.

Q5

Likelihood $x \mid \mu \sim \mathcal{N}(\mu, \sigma^2 = 4)$; prior $\mu \sim \mathcal{N}(\mu_0 = 0, \tau^2 = 1)$. Up to a constant,

$$\log p(\mu \mid x) = -\frac{(x - \mu)^2}{2\sigma^2} - \frac{(\mu - \mu_0)^2}{2\tau^2}.$$

$$\frac{d}{d\mu}: \; \frac{x - \mu}{\sigma^2} - \frac{\mu - \mu_0}{\tau^2} = 0 \;\Rightarrow\; \hat{\mu}_{\text{MAP}} = \frac{\tau^2 x + \sigma^2 \mu_0}{\sigma^2 + \tau^2} = \frac{1 \cdot 5 + 4 \cdot 0}{4 + 1} = 1.$$

The second derivative $-(1/\sigma^2 + 1/\tau^2) < 0$ confirms a maximum. The prior has four times the precision of the single observation, so the estimate is pulled strongly from $x = 5$ toward 0.

Q6

Let $\mu = E[X]$, a constant. By definition and linearity of expectation,

$\mathrm{Var}(X) = E[(X - \mu)^2] = E[X^2 - 2\mu X + \mu^2] = E[X^2] - 2\mu E[X] + \mu^2 = E[X^2] - \mu^2 = E[X^2] - (E[X])^2$. ■

Part 3: Principal Component Analysis

Q1

Let X be a random vector in $\mathbb{R}^d$ with mean $\mu = E[X]$. Its **covariance matrix** is

$$\Sigma = E\big[(X - \mu)(X - \mu)^T\big], \qquad \Sigma_{jk} = \mathrm{Cov}(X_j, X_k),$$

a symmetric positive semidefinite matrix whose diagonal holds the variances. Let u be a fixed unit vector ($u^T u = 1$) and $Y = u^T X$.

1. Mean: $E[Y] = u^T E[X] = u^T \mu$ (linearity).
2. Centered projection: $Y - E[Y] = u^T(X - \mu)$.
3. $\mathrm{Var}(Y) = E\big[(u^T(X - \mu))^2\big]$.
4. A scalar equals its transpose, so $(u^T(X - \mu))^2 = u^T(X - \mu)(X - \mu)^T u$.
5. u is constant and comes out of the expectation: $\mathrm{Var}(u^T X) = u^T E[(X - \mu)(X - \mu)^T]\, u = u^T \Sigma u$. ■

For a data matrix, $\hat{\Sigma} = \frac{1}{n-1} X_c^T X_c$, where X_c is the mean-centered data. PCA maximizes $u^T \Sigma u$ subject to $u^T u = 1$. Setting the gradient of the Lagrangian $u^T \Sigma u - \lambda(u^T u - 1)$ to zero gives $\Sigma u = \lambda u$. The best direction is therefore the top eigenvector, and the variance along it is $u^T \Sigma u = \lambda$.

Q2

I generated $n = 500$ samples with `numpy.random.default_rng(0).multivariate_normal`, using mean $[2, -1]$ and covariance $\begin{bmatrix} 3 & 1.2 \\ 1.2 & 1 \end{bmatrix}$. The population correlation is $1.2/\sqrt{3 \cdot 1} \approx 0.69$, and the sample correlation is $r \approx 0.70$. The scatter plot (left panel below) shows a tilted ellipse. x_1 has the larger spread, larger x_1 goes with larger x_2, and the major axis is about 25° above the x_1-axis rather than along either coordinate axis. The two features are partly redundant, which is what PCA takes advantage of.

Q3

The implementation (`pca` in `Assignment1.py`) follows the derivation in Q1:

1. **Center:** $X_c = X - \bar{x}$, because Q1 measures variance around the mean ($X - \mu$).
2. **Covariance:** $\hat{\Sigma} = X_c^T X_c/(n-1)$, the sample version of $E[(X-\mu)(X-\mu)^T]$.
3. **Eigendecomposition:** `np.linalg.eigh` solves $\hat{\Sigma}u = \lambda u$ and returns real eigenvalues and orthonormal eigenvectors for the symmetric matrix.
4. **Select:** sort the eigenvalues in descending order. The k-th eigenvector is PCk, the variance along it is λ_k, and the scores $X_c W$ are the data in PC coordinates.

Results for the generated data:

	eigenvalue λ	direction u (sign arbitrary)	explained variance
PC1	3.458	$[0.907,\ 0.422]$	89.4%
PC2	0.411	$[-0.422,\ 0.907]$	10.6%

As a check, the eigenvalues match the squared singular values of X_c divided by $n-1$, and the PC scores are uncorrelated with variances λ_1 and λ_2. In the middle panel each PC is drawn through the mean out to $\pm 2\sqrt{\lambda}$ (two standard deviations). PC1 runs along the long axis of the cloud and PC2 along the short axis, perpendicular to PC1.

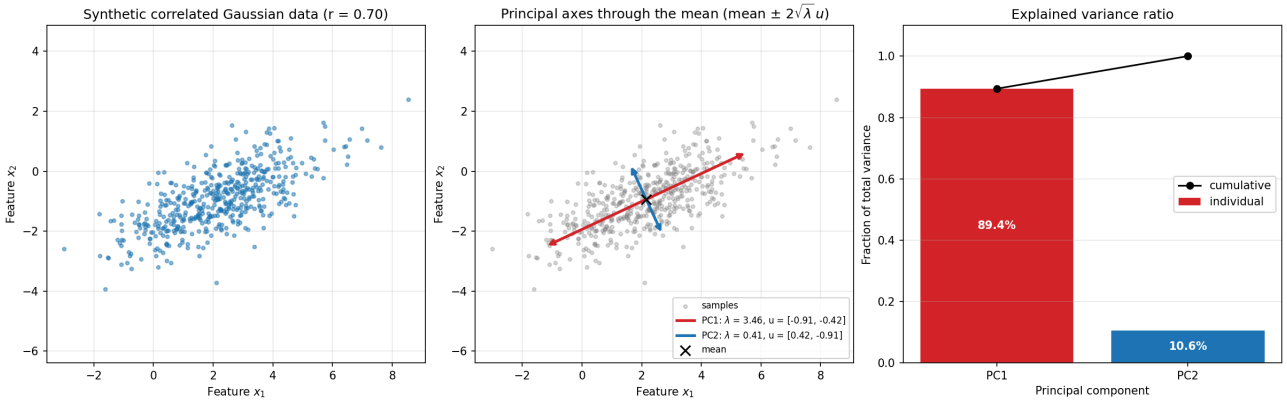


Figure 1. Left: synthetic correlated Gaussian data. Middle: principal axes through the mean. Right: explained variance ratio per component and cumulative.

Q4

The fraction of variance explained by PCk is $\lambda_k / \sum_j \lambda_j$, where $\sum_j \lambda_j = \mathrm{trace}(\hat{\Sigma})$ is the total variance. For this data, PC1 explains **89.4%** and PC2 explains **10.6%**. For the population covariance the values are $\lambda = 2 \pm \sqrt{2.44}$, giving 89.1% and 10.9%.

Discussion. Because the two features are strongly correlated, one direction captures almost 90% of the variability. Projecting onto PC1 alone reduces the data from 2D to 1D while keeping 89.4% of the variance. The 10.6% that is lost equals the mean squared reconstruction error, the narrow spread across the ellipse.

Trade-off. Keeping fewer components gives a smaller representation that is cheaper to store and compute with, less noisy, easier to visualize, and less affected by the curse of dimensionality. The cost is lost information, measured by the discarded variance. A common rule is to keep the smallest k whose cumulative explained variance passes a threshold such as 90–95%, or to cut at the elbow of the scree plot. Here, if 89% is enough, one component halves the dimension. If the features were uncorrelated with equal variance, each PC would explain 50%, and reducing dimension would lose as much information as it saves.